

## Study Guide for Population Growth and Demographics

This study guide provides examples for the topics that you might see on the first midterm from the lectures over population growth, and population demographics. This is NOT an exhaustive list of all content that might be covered on the exam. Use your notes from lecture, the lecture slides and video on Canvas, and to review all the topics that have been covered.

Dep → Density:

Anything worsens w/ more population.  
(competition, Disease predation.

1. Determine whether each of the scenarios below describes a **density-dependent** or **density-independent** control on butterfly populations

Dep a. The females lay many eggs on each plant, such that the caterpillars eat nearly all the leaves and 80% of the caterpillars starve before becoming adults

Too many for food.

Ind Dep b. The nursery plant species on which the butterflies lay their eggs is entirely removed from the habitat

Ind c. A severe rainstorm drowns 80% of the butterflies

Dep d. When butterflies are not abundant, predators switch to eating moths, which reduces the mortality rate of butterflies

Word = population based

Dep e. A contagious disease sweeps through the population, killing 80% of the butterflies

Ind f. A late spring cold snap reduces temperatures to outside the butterflies' tolerance and 80% of them die in their cocoons

2. You are studying the population of squirrels on the UC Davis campus. You capture 53 squirrels across campus and "tag" them by putting little UC Davis sweaters on them, which are super adorable. The next week, you survey campus and count a total of 123 squirrels, 31 of which are wearing sweaters. Given these data, what is your estimate for the size of the UC Davis squirrel population?

a. ~210 squirrels

b. ~129 squirrels

c. ~72 squirrels

d. ~437 squirrels

$$\frac{53}{x} = \frac{31}{123}$$

$$\frac{\text{Total marked}}{\text{Original Population}} = \frac{\text{Recaptured}}{\text{Second Sampling Size}}$$

$$x = \sim 210$$

$$31x = 53 \cdot 123$$

3. Below is a table for the size of a population of yeast cells in a flask. What is  $r$  for this population across this time window? Recall that the exponential growth equation is  $N_t = N_0 e^{rt}$

a. 0.747

b. 0.617

c. 0.768

d. 0.434

e. 0.482

f. 0.609

$$189.6 = 9e^{5r}$$

$$\ln \log\left(\frac{189.6}{9}\right) = 5r$$

$$r = \frac{\ln \log\left(\frac{189.6}{9}\right)}{5} = \frac{0.2647}{5} = 0.05294$$

| Hour | Yeast Cells per mL of water |
|------|-----------------------------|
| 0    | 9                           |
| 1    | 19                          |
| 2    | 35.2                        |
| 3    | 75.9                        |
| 4    | 117.1                       |
| 5    | 189.6                       |

Expo = Unlimited growth potential.

Natural Log!!!

4. The life table below shows data for a fictional population of otters living in the UC Davis Arboretum waterway. Use the table and the provided equations to calculate or estimate the metrics below.

Odds Count @ that age Number of offspring Age of Reproduction

| Age Class    | Age in Years | Proportion surviving ( $l_x$ ) | Age-specific fecundity ( $m_x$ ) | $l_x m_x$     | $x l_x m_x$    |
|--------------|--------------|--------------------------------|----------------------------------|---------------|----------------|
| Newborn      | 0            | 1.0                            | 0                                | 0             | 0              |
| Juvenile     | 1            | 0.97                           | 0                                | 0             | 0              |
| Young adult  | 3            | 0.91                           | 0.33                             | 0.3           | 0.9            |
| Adult        | 8            | 0.82                           | 2.07                             | 1.69          | 13.58          |
| Senior Adult | 14           | 0.43                           | 0.98                             | 0.42          | 5.89           |
| TOTALS       | ---          | ---                            | ---                              | $\Sigma 2.41$ | $\Sigma 20.37$ |

Averages

$l_x m_x$  = expected offspring @ that age

↳ 0.3 of newborns come from young adults

$x l_x m_x$  = how many offspring this age produces.

↳ young adults contribute 0.3 offspring; Because this happens at age 3, young adults contribute 0.9 towards average age of reproduction.

$R_0$  = avg # offspring

$G$  = avg age to reproduce

$r$  = growth speed per indiv

Fecundity = How many babies can you make?

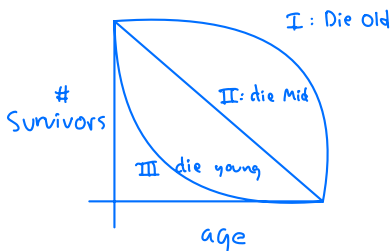
Net reproductive rate  
 $R_0 = \Sigma l_x m_x$

Generation time  
 $G = \frac{\Sigma x l_x m_x}{\Sigma l_x m_x}$

Intrinsic Growth Rate  
 $r \approx \frac{(\ln R_0)}{G}$

- a. the average number of offspring an otter will have throughout its lifetime = 2.41  
 b. the average age at reproduction = 20.37  
 c. the per capita rate of growth =  $\frac{\ln(2.41)}{20.37}$

5. Below is a life table for a fictitious population of turtles in the Arboretum on the UCD campus. Which of the statements below is best supported by the life table?

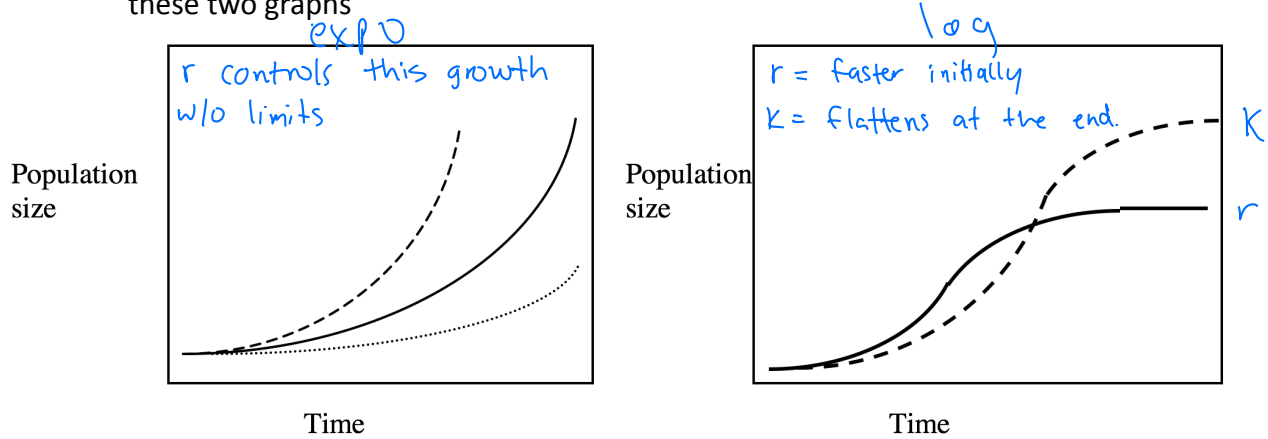


| Age class   | Age in Years | Proportion surviving ( $l_x$ ) | Age-specific fecundity ( $m_x$ ) | $l_x m_x$    | $x l_x m_x$  |
|-------------|--------------|--------------------------------|----------------------------------|--------------|--------------|
| hatchlings  | 0            | 1.0                            | 0                                | 0            | 0            |
| juvenile    | 1            | 0.25                           | 0                                | 0            | 0            |
| Young adult | ~3           | 0.20                           | 3.0                              | 0.6          | 1.8          |
| Adult       | ~5           | 0.10                           | 7.0                              | 0.7          | 3.5          |
| Old Adult   | 10+          | 0.10                           | 0                                | 0            | 0            |
| TOTALS      |              |                                |                                  | $\Sigma 1.3$ | $\Sigma 5.3$ |

1.5 0.5 = Type 3

- a. This population has a Type I survivorship curve  
 b. The age class with the highest mortality also has the highest fecundity adult + old adult share  
 c. On average, each individual in the population will have 5.3 offspring throughout its life  
 1.5 5.3 = age @ reproduction  
 d. Fecundity increases throughout the entire life of the individual True but drops to 0 @ 10+ years  
 e. None of the above are supported by this life table

6. Each graph below depicts population growth over time for different populations, shown by dotted, dashed, or solid curves. Answer the following questions using these two graphs



- Which of the graphs (left or right) describes exponential growth?
  - On the left graph, which population has the highest  $r$ ? *Dash*
  - On the right graph, which population has the highest  $K$ ? *Dash*
  - On the right graph, which population has the highest  $r$ ? *Solid*
7. In a population of birds, we find that as the number of birds increases, the birth rate is decreasing and the death rate is increasing. This population is likely...
- ~~a.~~ Experiencing exponential growth *read. ! Logistic ...*
  - ~~b.~~ Density independent *read*
  - ~~c.~~ At risk of extinction
  - d. Approaching its carrying capacity *Read carefully dipshit!!*
  - ~~e.~~ Maintaining a stable population size
8. Below are data for a fictitious population of ducks at the Arboretum on the UCD campus. Use the data to answer the following questions. (I've added in blank columns, which will help you in your calculations)

| Age class   | Average Age in Years ( $x$ ) | Proportion Surviving ( $l_x$ ) | Age-specific Fecundity ( $m_x$ ) | $Lxmx$ | $xlxmx$ |
|-------------|------------------------------|--------------------------------|----------------------------------|--------|---------|
| hatchlings  | 0                            | 1.0                            | 0                                | 0      | 0       |
| juvenile    | 1                            | 0.75                           | 0                                | 0      | 0       |
| Young adult | 5                            | 0.50                           | 10                               | 5      | 25      |
| Adult       | 7                            | 0.25                           | 12                               | 3      | 21      |
| Old Adult   | 10+                          | 0.01                           | 0                                | 0      | 0       |

- a. Calculate the following metrics for this population:

- The net reproductive rate ( $R_0$ ) =  $0 + 0 + 5 + 3 + 0 = 8$
- The generation time ( $G$ ) =  $0 + 0 + 25 + 21 + 0 = 46$
- The per capita (or intrinsic) rate of increase ( $r$ ) =  $\frac{\ln 8}{46} = 0.065$

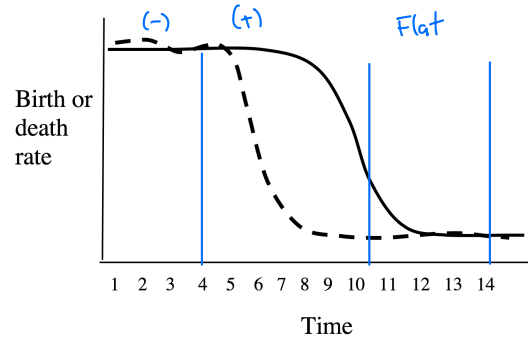
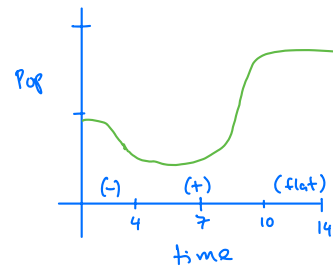
- b. Is this population increasing or decreasing? How do you know?

*Growing really slowly since  $r > 0$ .*

9. Assume the populations below are experiencing exponential growth. The exponential growth equation is  $N_t = N_0 e^{rt}$

- For a population of 275 individuals with  $r = 0.3$ , what will the population size be after 3 years?  $N_t = 275 e^{3 \cdot 0.3} = 676.391$
- For a population of 500 individuals with  $r = 0.04$ , what will the population size be after 3 years?  $N_t = 500 e^{0.04 \cdot 3} = 563.748$
- For a population of 275 individuals with  $r = 0.3$ , how many years will it take for the population to double in size?  $550 = 275 e^{0.3t} \rightarrow 2 = e^{0.3t} \rightarrow \frac{\ln(2)}{0.3} = t = 2.31 \text{ years}$
- How long will it take for a population with  $r = 0.04$  to double in size?  $2 = e^{0.04t} \rightarrow \frac{\ln(2)}{0.04} = t = 17.329 \text{ years}$

10. Given the following patterns of birth (solid line) and death (dashed line) rates in a population over time, sketch a graph of population size through time (hint; you can start at any non-zero population size).



11. Assume the populations below are experiencing logistic growth. Round your answers to the nearest whole organism. The logistic growth equation is

$$N_t = \frac{K}{1 + [(K - N_0)/N_0]e^{-rt}}$$

- For a population of 800 individuals with  $r = 0.15$ , living in an environment that can support 1000 individuals, what will the population size be after 1 year?  $K=1000, N_0=800, r=0.15, t=1 = 877.93$
- For a population of 200 individuals with  $r = 0.15$ , living in an environment that can support 1000 individuals, what will the population size be after 1 year?  $K=1000, N_0=200, r=0.15, t=1 = 225.08$
- Which of the populations from parts A and B above have a larger increase in population size in a single year? Explain why that population has a larger increase than the other population
- Is it possible biologically for a population to 'overshoot' the carrying capacity and have a larger population size than the environment can actually support? If so, what will happen to this population? Yes it's possible. The pop would likely

exp negative growth (limiting resources) declining back to K.

12. Explain the difference between  $r$  and  $R_0$ . Under which values of each is population growth positive, negative, and zero?  $R_0 = \text{avg \# of offspring a indiv of a species makes in life time}$

$r = \text{intrinsic growth rate} = \text{rate of whole population.}$   
 $r > 0 = \text{grow}$   
 $r = 0 = \text{stable}$   
 $r < 0 = \text{decline}$   
 $R_0 > 1 \rightarrow \text{replaces itself next gen + extra} \rightarrow \text{increase.}$   
 $R_0 < 1 \rightarrow \text{no replacement} \rightarrow \text{decrease.}$   
 $R_0 = 1 \rightarrow \text{replace itself} = \text{stable.}$

13. How can scientists estimate population sizes ( $N_0$ ) of naturally occurring organisms? How can we estimate per capita growth rates ( $r$ )?

$N_0 = \text{through mark recapture method}$

$r = \text{following a pop over time + count indivs.}$

or

Use life table analysis to estimate  $r$  from  $R_0$ .

Pop B increases more bc its  $N_0$  is further from K so it experiences less competition and faster growth rate.

$$N_t = \frac{K}{1 + [(K - N_0)/N_0]e^{-rt}} \rightarrow 200 = \frac{500}{1 + [0.1442]e^{-10r}} \rightarrow 200(1 + 0.1442e^{-10r}) = 500 \Rightarrow 1 + 0.1442e^{-10r} = \frac{5}{2} \rightarrow$$

$$0.1442e^{-10r} = \frac{3}{2} \rightarrow e^{-10r} = \frac{1.5}{.1442} \rightarrow -10r = \ln\left(\frac{1.5}{.1442}\right) \rightarrow r = \frac{\ln\left(\frac{1.5}{.1442}\right)}{-10} = -0.234 = r$$

$$K = 500, N_0 = 437, \\ N_t = 200, t = 10, r = ?$$

14. You are studying a population of feral cats which currently numbers 437 individuals. While the carrying capacity is approximately 500 cats for this region, the population is causing problems by hunting native birds and lizards and so your management goal is to get the population of cats down to 200 cats in the next 10 years.

a. What  $r$  value would you need to achieve in order to hit this target?

$$r \approx \frac{(\ln R_0)}{G} \quad rG = \ln R_0$$

$$r(1) = \ln R_0$$

$$e^r = R_0 \quad e^{-0.234} = R_0 \rightarrow 0.791$$

b. How many offspring, on average, can cats in this population have throughout their lifetime in order to stay below that  $r$  value? Assume generation time in cats is about 1 year.

15. You are studying a population of geckos in Hawai'i: you tag 15 geckos with unique paint markings and release them back into the habitat, then return a week later. On your second visit, you observe 40 geckos in the habitat, none of which bear a paint tag. Propose three explanations for why your mark-recapture studied failed.

1) Tags don't last / aren't detectable } we think none are recaptured  
 • Molting Lizards / snakes  
 • Tags fall off / self-removed  
 = pop looks huge

2) Marked Indivs die / get eaten } Marked groups disappear  
 • Tags make indivs more obv for predation  
 = no recapture

3) Immigrate / Emigrate } Now our sample changes  
 • Mark indivs leave that habitat  
 • New unmarked ones take over  
 population completely.

4)

Mark/Cap Assumptions:

- 1) Closed population
- 2) tags persist
- 3) Equal capture prob
- 4)